MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PAPER-DIELECTRIC, DIRECT CURRENT (HERMETICALLY SEALED IN METAL CASES), STYLE CP80

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-25.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DO NOT USE FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY

REQUIREMENTS:

Dimensions: See figure 1 and table I.

Case: Magnetic or nonmagnetic material, uninsulated.

Design and construction: Capacitors shall be of the design, construction and physical dimensions as specified in figure 1 and table I.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC voltage rating: See table I.

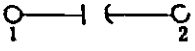
Operating temperature range:
-55° to +85° C.

Vibration grade: 1 (10 to 55 Hz, incl.)

Terminal:

Symbol C - - - - Threaded stud and nuts.

Circuit diagram:

Symbol 1 - - - - 

Voltage derating with temperature:
See figure 2.

Dielectric withstanding voltage: Method 301 of MIL-STD-202.
Terminal to terminal - As specified in table II.
Terminal to case (when case is not a terminal) - As specified in table II. Capacitors shall withstand specified voltages with no permanent damage or open- or short-circuiting.

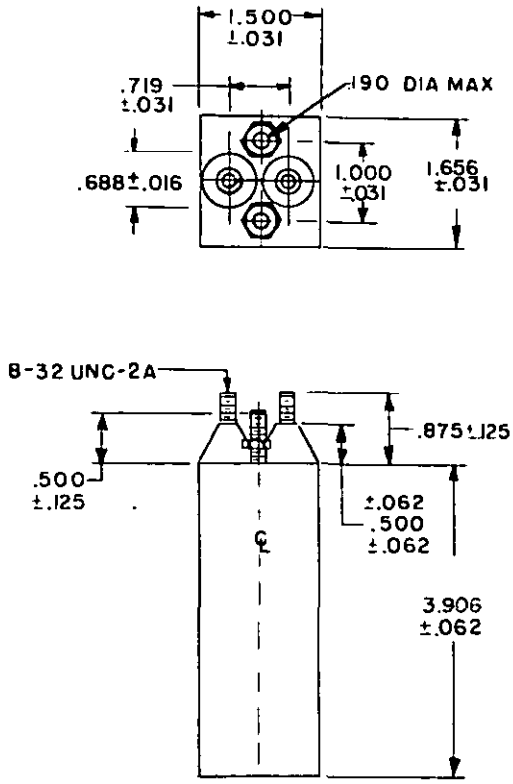
Insulation resistance: Method 302 of MIL-STD-202.
At 25° C:
Terminal to terminal - As specified in table III.
Terminal to case (when case is not a terminal) - Shall exceed 3,000 megohms.
At 85° C:
Terminal to terminal - As specified in table III.

Capacitance:
Method 305 of MIL-STD-202.
Measured capacitance shall be within the tolerance specified in table I.

Dissipation factor:
When measured as specified in MIL-C-25, shall not exceed 1 percent.

FSC 5910

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INCHES	MM
.016	.41
.031	.79
.062	1.57
.125	3.18
.190	4.83
.500	12.70
.688	17.48
.719	18.26
.875	22.23
1.000	25.40
1.500	38.10
1.656	42.06
3.906	99.13

NOTES:
1. Dimensions are in inches.
2. Mounting hardware (including lockwasher) supplied with capacitor.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CP80 capacitor.

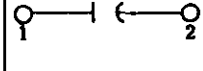
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TABLE I. Style CP80 capacitor.

Type designation.	DC voltage rating (see fig. 2)	Capacitance	Capacitance tolerance 1/	Characteristic	Vibration grade
	Volts	μf			
CP80C1BE405L1	400	4	L	B	1
CP80C1BE205L1	400	2	L	B	1
CP80C1BG105L1	1,000	1	L	B	1

1/ Capacitance tolerance in percent: L ± 15.

TABLE II. Dielectric withstanding voltage test.

Circuit symbol	Circuit diagram	Test	Test connections	Test voltage (percent of rated dc voltage)	Time test applied (minutes)
1		Terminal to terminal	1 to 2	200 1/	1 2/
		Terminal to case	1 and 2 to case	400 5/	1 3/ 4/
				200 + 1,000 6/ volts	1 3/ 4/

1/ 175 percent of rated dc voltage after temperature-and immersion-cycling and moisture resistance tests.

2/ For quality conformance inspection and at the option of the supplier, either 200 percent of rated dc voltage shall be applied for 15 seconds or 250 percent of rated dc voltage for not less than 1 second.

3/ For quality conformance inspection, the period of test-voltage application shall be 1 second.

4/ For quality conformance inspection, applications may be made between each terminal individually and the case, at the option of the supplier.

5/ For capacitors having dc voltage ratings of 600 volts and less.

6/ For capacitors having dc voltage ratings greater than 600 volts.

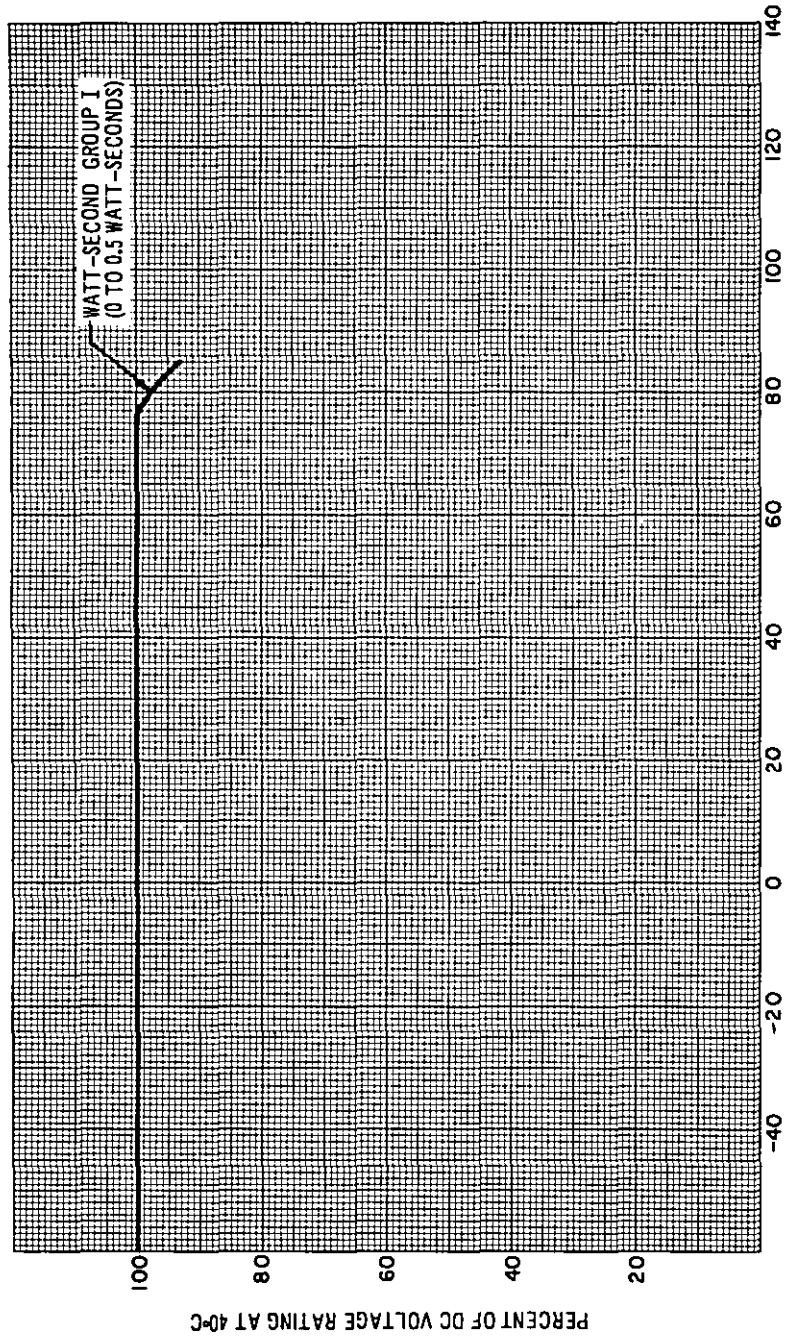
TABLE III. Terminal to terminal insulation resistance measurements.

Test temperature	Minimum insulation resistance 1/ Characteristic B
At 25° C - - - - - - - - - - -	1, 500 megohm-microfarads 2/
At 85° C - - - - - - - - - - -	15 megohm-microfarads 2/

1/ A potential equal to the dc rated voltage or 500 volts dc, whichever is less, shall be applied.

2/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

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NOTE: Permissible operating voltages, while based on incomplete data, are the capacitor suppliers' best estimate to provide a life expectancy of 8,800 hours of continuous operation at higher ambient temperatures. Longer life can be expected by operation at voltages lower than indicated on the curve; for example, a life expectancy of 44,000 hours may be obtained by operation at 70 percent of the voltage determined by use of this curve. Also, a life longer than 8,800 hours may be expected at the voltage determined by use of this curve if the high ambient temperature prevails for only a portion of the whole operating time.

FIGURE 2. Voltage derating for ambient temperature (characteristic B).

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Barometric pressure (flashover):
Method 105 of MIL-STD-202.
Test condition - B (50,000 feet).
Test potential - 125 percent of rated dc voltage, applied for a minimum of 1 minute, without visible damage, external flashover, or open- or short-circuiting.

Vibration (grade 1, 10 to 55 Hz, incl):
Method 201 of MIL-STD-202.
Duration - 3 hours (60 minutes in each direction).
Measurements - An electrical signal of 1±0.2 kHz, at a level of 1±0.5 volts applied during the last 30 minutes in each direction. There shall be no momentary arcing or breakdowns, open- or short-circuiting, or mechanical damage.

Salt spray (corrosion): Method 101 of MIL-STD-202, test condition B (48 hours), with no evidence of corrosion or obliteration of marking.

Temperature cycling: Method 102 of MIL-STD-202, test condition D.

Immersion: Method 104 of MIL-STD-202, test condition C, except that the duration of each immersion shall be 30 minutes, and change between solutions shall be made in not more than 3 seconds.
Measurements after final cycle:
Dielectric withstanding voltage, at 25° C:
Terminal to terminal - As specified in table II.
Terminal to case (when case is not a terminal)- As specified in table II.
Insulation resistance at 25° C:
Terminal to terminal - Not less than 30 percent of the value specified in table III.
Terminal to case (when the case is not a terminal) - Not less than 1,500 megohms.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and E:
Test condition A (pull) - Applied force - 5 pounds.
Test condition E (torque) - 2 pound-inches.

Moisture resistance: Method 106 of MIL-STD-202.
Polarization voltage - 100 volts.
Final measurements:
Dielectric withstanding voltage and insulation resistance requirements - As specified in immersion.

Seal:
Method 112 of MIL-STD-202, test condition A, except that bath temperature shall be $+85^{\circ}\ ^{+3}_{-0}$° C.
Liquid-filled capacitors - 10 minutes on each of two opposite sides at $+85\ ^{+3}_{-0}$° C.
There shall be no evidence of leakage.

Low temperature and capacitance change with temperature:
Low temperature -
Conditions:
-55° ±3° C for 48 ±4 hours with rated dc voltage applied.

Capacitance change with temperature:
Not to exceed +10, -30 percent.

Life:
Method 108 of MIL-STD-202, test condition B ($250\ ^{+8}_{-0}$ hours).
Electrical load: 120 percent of rated dc voltage.
Final measurements (at 25° C):
Insulation resistance:
Terminal to terminal: Not less than 30 percent of the value specified in table III.
Capacitance:
Within 10 percent of initial value.
Dissipation factor:
Change not more than 1 percent from initial value.

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Inspection lot: An inspection lot may consist of capacitors described herein and one or more of styles CP81 and CP82.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army -
Navy - MC
Air Force - 19

Preparing activity:
Air Force - 11

(Project 5910-0824-13)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R001

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

PLATE NO 15413

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